Information Retrieval in the Medical Domain: A Comparative Study on TREC-COVID

Hands on NLP

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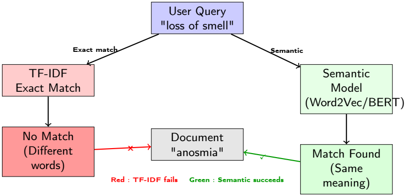
12 février 2026

Context :

- TREC-COVID challenge
- 171,000+ scientific papers

Limitations :

- Exact-match systems (TF-IDF) fail
- Poor retrieval performance
- Semantic gap



Approaches

To solve the problem, we have used different approaches, that we will present to you :

- **Baseline :** TF-IDF (Term Frequency-Inverse Document Fre- quency)
- **Static Embeddings** : Word2Vec trained on the corpus vs. Pre-trained BioWord2Vec.
- **Neural Models** : Deep Averaging Network (DAN) with triplet loss.
- **Transformer Models :** BioBERT (standard) and Sentence- BERT (optimized for semantic similarity)

Dataset and Methodology

TREC-Covid

The dataset we used is the TREC-COVID, a dataset created specifically as an information retrieval task on Kaggle in 2020. The creators of the dataset, a group of researchers from the NIST published in 2021 an article describing the dataset and the task : **Searching for Scientific Evidence in a Pandemic: An Overview of TREC-COVID**

It contains :

- **Corpus** : 171,332 scientific articles from the CORD-19 collection, each containing a title and text (abstract or full content).
- **Queries** : 50 distinct topics in natural language, representing real-world information needs.
- **Qrels** : Expert relevance judgments used as Ground Truth, with relevance scores on a -1-2 scale (-1 : Document not evaluate, 0 : not relevant, 1 : partially relevant, 2 : highly relevant). They are used to evaluate a model, by comparing the relevance predicted by the model to the relevance the experts pointed out.

Models

TF-IDF

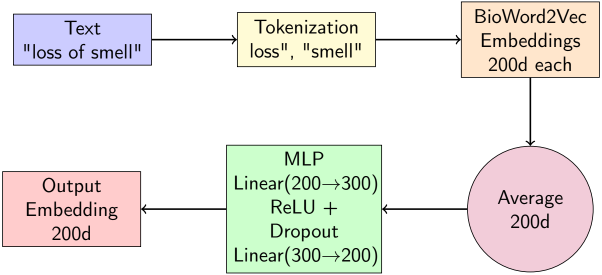
The first model we use is a TF-IDF, implemented using the scikit-learn TfidfVectorizer with default parameters.

Word2Vec Variants.

1. **W2V-General :** Trained with Gensim's Word2Vec. Parameters : 200-dimensional embeddings, window size 5, minimum word count 2, trained for 5 epochs.
2. **BioWord2Vec** : Pre-trained model trained on 1.5 million biomedical words. Does not use our dataset.
3. **BioWord2Vec Fine-tuned :** The pre-trained BioWord2Vec model further fine-tuned on TREC-COVID for 5 epochs.

For all Word2Vec variants, documents are represented by averaging word embedding, combining title and text.

Models



Bi-encoder architecture : Queries and documents share the same encoder.
Training data : Triplets (query, positive document, negative document) encode semantic relationships.

Transformer Models

Models

1. **BioBERT (Standard) :** We used the *dmis-lab/biobert-base-cased-v1.1* model from Hugging Face. Documents and queries are tokenized with a maximum length of 512 tokens. We extract the [CLS] token embedding (768 dimensions) as the document/query representation. No fine-tuning was performed for the retrieval task.
2. **Sentence-BERT (S-PubMedBert) :** We used *pritamdeka/S-PubMedBert-MS*, a Siamese BERT network fine-tuned on MS-MARCO (general IR) and PubMed (biomedical domain).

Key Difference : Sentence-BERT embeddings are optimized for semantic similarity through contrastive learning on pairs of similar/dissimilar texts. BioBERT's [CLS] token is trained for classification tasks, not similarity search.

Evaluation and Performance Comparison

- **Evaluation Metric :** NDCG@10 (Normalized Discounted Cumulative Gain at 10)
  - Prioritizes highly relevant documents at the top of the ranking list.
  - Accounts for both relevance and ranking position, making it ideal for IR evaluation.
  - Reported : Mean NDCG@10 across all 50 queries, with standard deviation, minimum, and maximum scores.

Model	**Mean**	**Std**	**Min**	**Max**
TF-IDF (Baseline)	0.5868	0.3350	0.0000	1.0000
Word2Vec (Corpus)	0.6099	0.3301	0.0000	1.0000
BioWord2Vec (Pre-trained)	0.7366	0.3029	0.0000	1.0000
BioWord2Vec (Fine-tuned)	0.6259	0.3263	0.0000	1.0000
DAN	0.6630	0.2696	0.0000	1.0000
BioBERT (Standard)	0.2886	0.3454	0.0000	1.0000
Sentence-BERT	**0.8171**	**0.2305**	**0.0000**	**0.9986**

Conclusion

Best Model

This study evaluated multiple IR strategies for the TREC-COVID challenge, ranging from classical lexical methods to state-of-the-art neural architectures. We demonstrated that Sentence-BERT is the most effective approach (NDCG@10 : 0.8171), leveraging both domain-specific pretraining and an architecture optimized for semantic similarity.

Key Findings

1. **Domain-specific pre-training is crucial :** BioWord2Vec (+25.5%) and Sentence-BERT (+39.2%) significantly outperform generic models.
2. **Task-specific optimization matters :** Sentence-BERT's similarity-focused training outperforms BioBERT's classification focused embeddings by 183%.
3. **Model complexity requires data :** Trainable models (DAN, fine-tuned W2V) underperform simpler approaches when training data is limited (50 queries).
4. **Simple can be effective :** BioWord2Vec averaging (0.7366) outperforms complex neural architectures (DAN : 0.6630) in low-data scenarios.

Thanks you for your attention !

Questions ?